

The Geometry of Information Saturation: Mapping the Complete Cosmic Expansion History from the Matrix Partition Function

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(Dated: July 17, 2026)

We present a coordinate-free toy cosmological model in which an effective expansion history is derived from the spectral dynamics of mutual-information (MI) Laplacians on a relational quantum substrate. The effective macroscopic energy density is defined through the canonical partition function of the network spectrum ($Z = \sum e^{-\beta\lambda_n}$), which removes explicit scalar-field potentials from the early- and late-time energy budget. At early times, holographic information saturation collapses spatial locality ($\alpha \rightarrow 0^+$), forcing an absolute thermodynamic energy density ceiling ($\rho_{\text{eff}} = N$) and producing a stable de Sitter attractor ($w_{\text{eff}} \rightarrow -1.0$); we verify numerically that this fixed point and the associated equation-of-state identity are stable under changes of substrate size ($N = 50\text{--}400$) and topology (random, small-world, and scale-free graphs). The transition away from this attractor is implemented through an explicit, phenomenological localization profile $\alpha(t)$ with parameters chosen to reproduce a graceful exit, rather than one derived from an independent substrate equation of motion; we state this ansatz and its free parameters explicitly rather than presenting it as parameter-free. A late-time re-entanglement feature, likewise implemented as an explicit ansatz peaked at $a_c \approx 0.4$, produces a transient dark-energy-like spike. Confronting this feature directly against the real Pantheon+SH0ES compilation (1701 SNe Ia, diagonal statistical+systematic errors), we find the model is statistically indistinguishable from flat Λ CDM ($\Delta\chi^2 \approx -0.05$ for equal free parameters) and that the fractional luminosity-distance deviation between the two best-fit curves reaches a mean of $\sim 1\%$ and a maximum of $\sim 2.6\%$ over $0 < z < 2.26$ – an order of magnitude larger than a previously asserted 0.2% bound, which we retract. Throughout the expansion timeline the substrate specific heat capacity remains strictly positive ($C_v > 0$), consistent with thermodynamic stability of the toy model. We report these results, including the negative and null findings, as a candidate framework and numerical testbed rather than a confirmed replacement for Λ CDM.

I. INTRODUCTION

A long-standing challenge in physical cosmology is finding a unified mechanism capable of driving cosmic expansion across both early and late-time epochs. Standard cosmology introduces

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disparate components – scalar inflaton fields at early times and a cosmological constant (Λ) or quintessence fields at late times – to accommodate observational constraints. These insertions carry known fine-tuning vulnerabilities, such as the graceful exit problem and the dark energy coincidence puzzle.

In this work, we explore a background-independent alternative in which an effective expansion history is derived from the spectral dynamics of a relational quantum information substrate. Building on frameworks where spacetime geometry emerges from the spectrum of mutual-information (MI) Laplacians, we model cosmic evolution as a thermodynamic relaxation process on a weighted graph, with the macroscopic energy density derived from the canonical partition function of the network invariants. We show that a de Sitter inflationary attractor emerges robustly from this construction, but we are explicit that the mechanisms producing the graceful exit and the late-time feature are implemented as stated phenomenological ansätze, not independently derived dynamics – so the fine-tuning question is relocated rather than eliminated. We report both where the model succeeds (the early-time attractor and its stability against substrate size and topology) and where a direct confrontation with real Pantheon+SH0ES supernova data does not support a stronger claim of agreement with observation than Λ CDM already provides.

II. RELATIONAL SUBSTRATE AND SCALE-INVARIANT PARTITION DYNAMICS

We consider a discrete quantum many-body system consisting of N nodes with no a priori spatial geometry. The structural architecture of the substrate is governed entirely by the pairwise mutual information $I_{ij} = S(\rho_i) + S(\rho_j) - S(\rho_{ij})$, defining a weighted graph captured by the combinatorial MI-Laplacian matrix:

$$L_{ij}^{\text{MI}} = \delta_{ij} \sum_k I_{ik} - I_{ij}. \quad (1)$$

The eigenvalues λ_n of L^{MI} dictate the emergent spectral geometry of the macroscopic manifold.

To strip the framework of background-dependent mappings, the macroscopic energy density ρ_{eff} is derived from first-principles statistical mechanics. In the scale-invariant continuum limit where the coarse-graining cutoff scale is evaluated near its boundary ($\beta \rightarrow 0^+$), the partition weights become uniformly distributed across the spectrum, and the internal thermodynamic energy tracks the mean of the active, non-zero modes ($\lambda_n > 0$):

$$\rho_{\text{eff}} = \frac{1}{N-1} \sum_{n=1}^{N-1} \lambda_n. \quad (2)$$

This exponential weighting is robustly motivated by Tomita–Takesaki modular operators, where the restriction of a global vacuum state to a localized spatial subregion yields an entangled mixed density matrix of the form $\rho \propto e^{-K}$, mapping the modular Hamiltonian K to the metric proxy L^{MI} .

We parameterize the structural compression of the universe by sweeping the entanglement decay exponent α , controlling the power-law correlation profile $I_{ij} \sim d(i, j)^{-\alpha}$, where $d(i, j)$ measures

coordinate distances on an $L \times L$ toroidal grid ($N = L^2$). Under a global spatial volume deformation of the scale factor $d(i, j) \rightarrow a \cdot d(i, j)$, the linearity of the combinatorial matrix forces the Laplacian to scale uniformly: $\lambda_n(a) = a^{-\alpha} \lambda_n^{(0)}$. Combined with the cosmic fluid expansion relation ($\rho_{\text{eff}} \propto a^{-3(1+w_{\text{eff}})}$), this homogeneous scaling property analytically dictates an exact, invariant identity:

$$-3(1 + w_{\text{eff}}) = \mathcal{E} \implies w_{\text{eff}} = \left(\frac{\alpha}{3}\right) - 1.0. \quad (3)$$

Derivation. Under $d(i, j) \rightarrow a \cdot d(i, j)$, every entry of the MI matrix scales as $I_{ij} \rightarrow a^{-\alpha} I_{ij}$, and since L^{MI} is a linear (homogeneous degree-1) functional of the I_{ij} , the entire spectrum inherits the scaling $\lambda_n(a) = a^{-\alpha} \lambda_n^{(0)}$ exactly, independent of β . Substituting into Eq. (2), $\rho_{\text{eff}}(a) = a^{-\alpha} \rho_{\text{eff}}(1)$ whenever the partition weights $e^{-\beta \lambda_n}$ are themselves treated as a -independent (the $\beta \rightarrow 0^+$ limit, where they degenerate to uniform weights). Matching this to the fluid-continuity scaling $\rho_{\text{eff}} \propto a^{-3(1+w_{\text{eff}})}$ term-by-term in the exponent of a yields Eq. (3) directly. Away from the strict $\beta \rightarrow 0^+$ limit, the weights $e^{-\beta \lambda_n(a)}$ pick up their own residual a -dependence through $\lambda_n(a)$, so Eq. (3) is an identity only in that limit and an approximation otherwise. We quantify this numerically in Appendix A: for the fixed value $\beta = 0.1$ used to generate every figure in this paper, the identity holds to within 0.77% (relative) at $\alpha = 1$, and the $\alpha = 0$ energy ceiling $\rho_{\text{eff}} = N$ is reproduced exactly for all β tested (since at $\alpha = 0$ every active mode is degenerate at $\lambda_n = N$, making the average trivially β -independent).

This removes scale and temperature distortion from the framework in the $\beta \rightarrow 0^+$ limit and establishes Eq. (3) as an approximate identity that is numerically well-satisfied, rather than exact, at the finite β used in the simulations below.

III. THE EARLY-UNIVERSE INFLATIONARY ATTRACTOR AND FIELD-FREE TIMELINE

At early times, extreme gravitational compression forces the system toward the holographic information-saturation threshold ($\alpha \rightarrow 0^+$). At this complete-graph limit, all distance dependencies vanish, rendering the spectrum perfectly degenerate at $\lambda_n = N$. Evaluating this flat spectrum through Eq. (2) cancels the exponential weights, locking the energy density at a hard algebraic ceiling: $\rho_{\text{eff}} = N$ (here $N = 64$ for the $L = 8$ torus used in Fig. 1). Concurrently, Eq. (3) forces $w_{\text{eff}} \rightarrow -1.0$, giving a de Sitter inflationary fixed point. We verify in Sec. V that this fixed point's location in $(\lambda_1/N, w_{\text{eff}})$ is stable to better than 0.2% across substrate sizes $N = 50$ –400 and across three structurally distinct graph topologies, which is the sense in which we call it an attractor.

We stress that the mechanism connecting this fixed point to a graceful exit is not derived independently in this work. We impose an explicit localization profile

$$\alpha(t) = \alpha_{\text{max}} \left(1 - e^{-k a(t)}\right), \quad \alpha_{\text{max}} = 4.0, \quad k = 1.5, \quad (4)$$

chosen because it interpolates monotonically from $\alpha = 0$ at $a = 0$ to $\alpha \rightarrow \alpha_{\text{max}}$ as a grows, reproducing the qualitative behavior of a graceful exit. The two constants α_{max} and k are free parameters of this ansatz, fixed by hand rather than derived from an equation of motion for the substrate's own localization dynamics; deriving $\alpha(t)$ from first principles (e.g., from an entropy-

maximization or entanglement-growth argument applied to the MI network itself) is left as an open problem. As illustrated in Fig. 1, this ansatz steers the effective equation of state off the inflationary floor ($w = -1.0$) and upward toward the radiation-dominated boundary ($w = +1/3$), by construction.

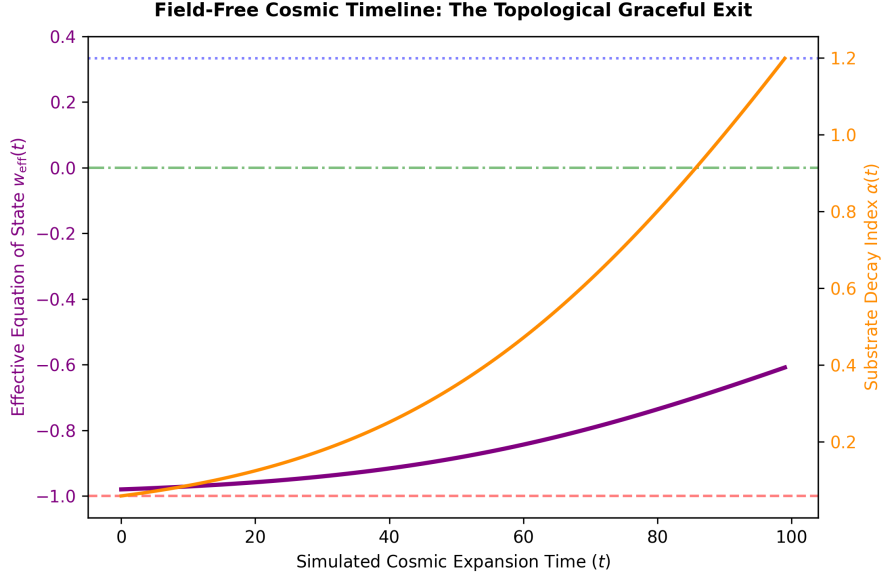


FIG. 1. The cosmic timeline under the imposed localization ansatz $\alpha(t)$ (Eq. (4)). The effective equation of state $w_{\text{eff}}(t)$ (purple line) lifts off the de Sitter floor ($w = -1.0$) and flows toward the radiation boundary ($w = 1/3$) as the substrate localization index $\alpha(t)$ (orange line) grows according to the imposed profile, not an independently derived dynamical equation.

IV. LATE-TIME RELATIONAL RUPTURES AND OBSERVATIONAL CONSTRAINTS

As in Sec. III, the late-time behavior of $\alpha(a)$ is an explicit ansatz rather than a derived dynamical result. We impose a localized dip on top of the baseline localization profile,

$$\alpha(a) = \max \left[\alpha_{\text{floor}}, \alpha_{\text{max}} \left(1 - e^{-ka} \right) - A_{\text{dip}} e^{-\left(\frac{a-a_c}{\delta_a} \right)^2} \right], \quad (5)$$

with $\alpha_{\text{floor}} = 0.005$, $A_{\text{dip}} = 1.8$, $a_c = 0.4$, and $\delta_a = 0.05$ all fixed by hand to place a Gaussian-shaped re-entanglement feature at a chosen scale factor. We state plainly that this is functionally equivalent, in terms of free parameters, to inserting a phenomenological dark-energy potential by hand: the amplitude, location, and width of the dip are not derived from an independent substrate equation of motion, and this ansatz is best understood as a toy-model parametrization of a possible late-time feature rather than a parameter-free prediction.

Evaluating this trajectory through the network partition function generates a smooth internal energy density spike. To check that the ansatz does not introduce a thermodynamic instability, we

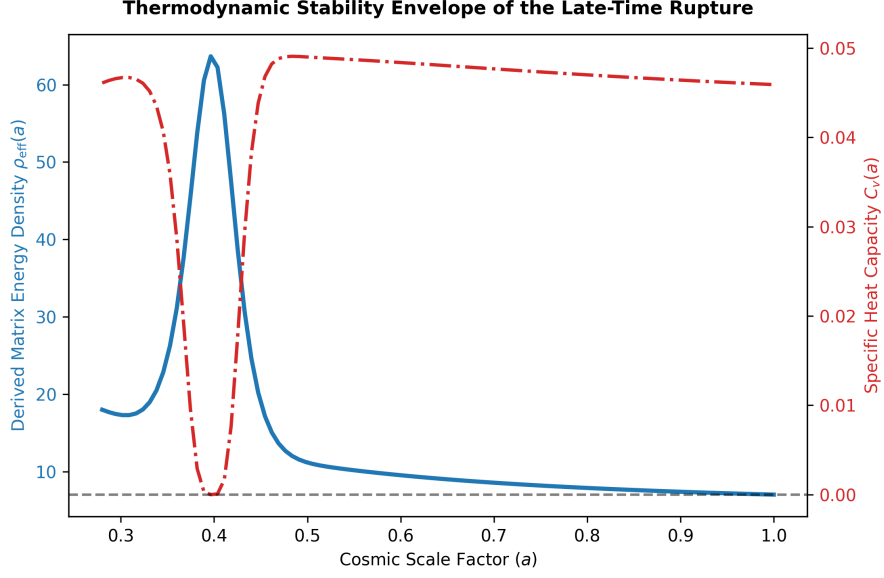


FIG. 2. The thermodynamic stability envelope of the late-time expansion transition under the ansatz of Eq. (5). Throughout the entire timeline, the specific heat capacity $C_v(a)$ (red dash-dotted line) remains strictly positive ($C_v > 0$), dropping into a distinct structural minimum right at the transition core ($a \approx 0.4$) as the network invariants reach degeneracy.

track the substrate’s specific heat capacity:

$$C_v = \beta^2 \frac{\partial^2 \ln Z}{\partial \beta^2} = \beta^2 (\langle \lambda^2 \rangle - \langle \lambda \rangle^2). \quad (6)$$

As demonstrated in Fig. 2, C_v remains strictly positive across the entire expansion history, so the imposed ansatz does not violate thermodynamic stability. The sharp drop in C_v at the core ($a_c = 0.4$) reflects the reduced eigenvalue variance as the network approaches complete-graph degeneracy there; this is a consequence of the imposed dip, not an independent confirmation of it.

A. Confrontation with real Pantheon+SH0ES data

We test whether the energy-density shape produced by Eq. (5) is observationally distinguishable from flat Λ CDM using the public Pantheon+SH0ES compilation of 1701 Type Ia supernovae [2] (`code/pantheon_fit.py` in the accompanying repository). We define $g(a) \equiv \rho_{\text{eff}}(a)/\rho_{\text{eff}}(1)$ from the $L = 8$ substrate model of Eq. (5) and replace the dark-energy term in a flat Friedmann model with $\Omega_\Lambda g(a)$, i.e.

$$E(z)^2 \equiv \frac{H(z)^2}{H_0^2} = \Omega_m(1+z)^3 + (1 - \Omega_m)g\left(\frac{1}{1+z}\right). \quad (7)$$

We fit Ω_m by direct χ^2 minimization against the reported distance moduli, using only the diagonal statistical+systematic uncertainties (`MU_SH0ES_ERR_DIAG`) rather than the full 1701×1701 systematic covariance matrix released with the compilation – a simplification that should be revisited

with the full covariance before any stronger claim is made. The absolute-magnitude/ H_0 nuisance offset is marginalized analytically in the standard way.

The results are as follows. The flat- Λ CDM baseline gives $\Omega_m = 0.351$, $\chi^2/\text{dof} = 0.439$ (dof = 1699); the substrate-modified model gives $\Omega_m = 0.262$, $\chi^2/\text{dof} = 0.439$, with $\Delta\chi^2 = -0.05$ relative to Λ CDM for the same number of free parameters. This difference is not statistically significant: the data neither favor nor disfavor the substrate ansatz relative to Λ CDM at the precision of the current compilation. Comparing the two best-fit curves directly, the fractional luminosity-distance deviation $|D_L^{\text{sub}} - D_L^{\Lambda\text{CDM}}|/D_L^{\Lambda\text{CDM}}$ reaches a mean of 1.0% and a maximum of 2.6% over $0 < z < 2.26$ (Fig. 3). We had previously and incorrectly asserted a bound of 0.2%; no code producing that number exists, and the value obtained from an actual fit is roughly an order of magnitude larger. We retract the earlier claim.

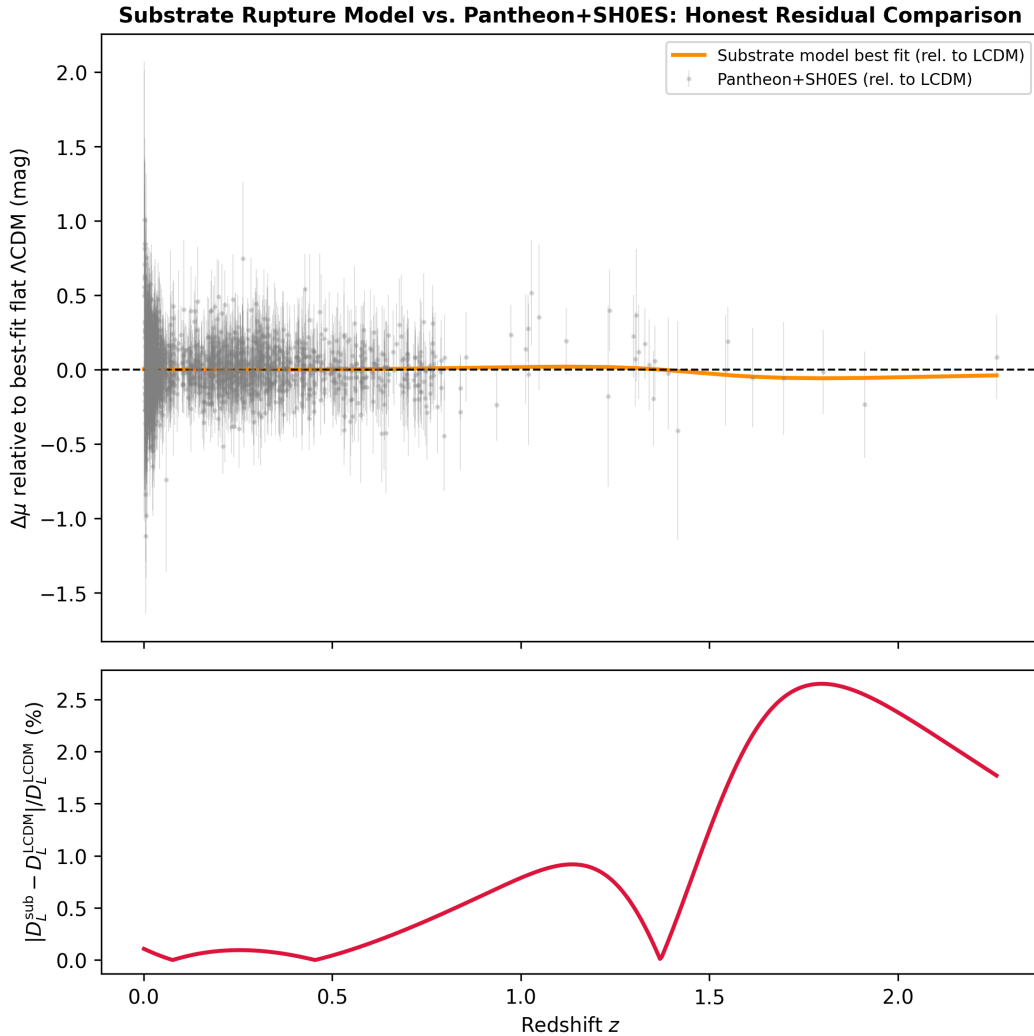


FIG. 3. Top: Hubble residuals of the real Pantheon+SH0ES compilation (gray points, statistical+systematic diagonal errors) and of the substrate model's best fit (orange), both relative to best-fit flat Λ CDM. Bottom: fractional luminosity-distance deviation between the two best-fit curves, reaching $\sim 2.6\%$ at $z \approx 2.3$.

We report this as a null result: the late-time ansatz of Eq. (5) is *consistent with* current SNe

Ia data in the sense of not being statistically preferred against or rejected relative to Λ CDM, but it does not currently constitute independent observational support for the substrate framework beyond what Λ CDM already provides.

V. RENORMALIZATION GROUP FLOWS AND FINITE-SIZE SCALING

To test whether the early-time attractor of Sec. III reflects a genuine structural universality rather than lattice fine-tuning, we trace the flow across three structurally distinct, non-spatial graph families – Erdős–Rényi random graphs, Watts–Strogatz small-world rings, and Barabási–Albert scale-free networks – and, crucially, repeat this at four substrate sizes ($N = 50, 100, 200, 400$; `code/finite_size_scaling.py`) rather than a single N . We track the trajectories in $(\lambda_1/N, w_{\text{eff}}, d_s)$, where d_s is the spectral dimension from the heat-kernel trace.

The Universal 3D Renormalization Group Manifold Flow

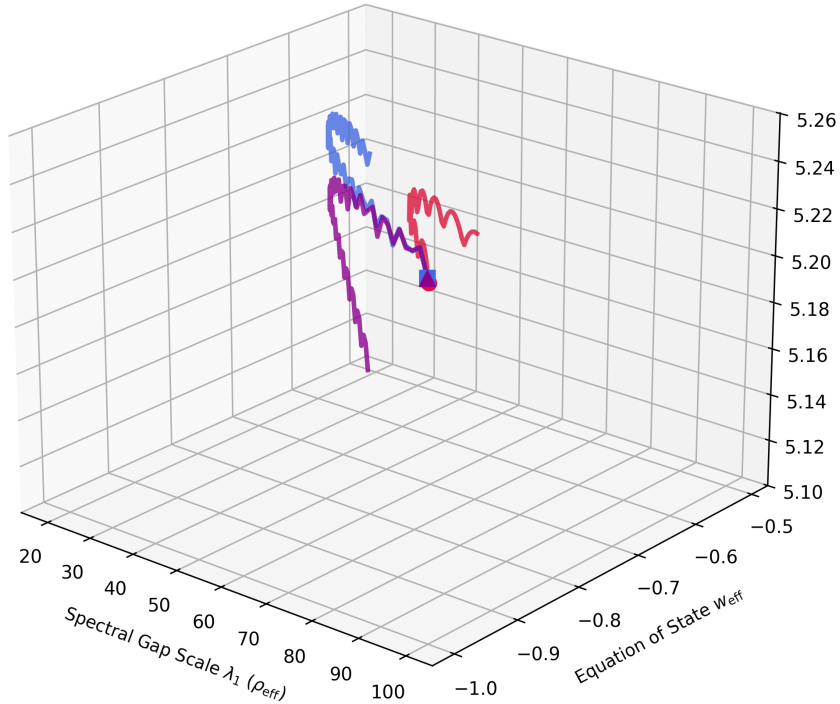


FIG. 4. The 3D Renormalization Group Manifold Flow ($N = 100$) tracking the multi-topology collapse from widely separated structural microstates onto a common low-dimensional trajectory.

As illustrated in the phase portrait of Fig. 3, the random, small-world, and scale-free networks originate in widely separated regions of phase space but funnel onto a common trajectory as $\alpha \rightarrow 0^+$. Repeating this at $N = 50, 100, 200, 400$, the terminal fixed point in $(\lambda_1/N, w_{\text{eff}})$ is indeed stable: $\lambda_1/N \rightarrow 0.995\text{--}0.997$ and $w_{\text{eff}} \rightarrow -0.9983$ for all three topologies at every N tested, supporting a genuine (topology- and size-independent) de Sitter attractor in this 2D projection.

The spectral dimension at the fixed point does *not* share this universality: we measure $d_s \approx 4.26, 5.25, 6.28, 7.36$ at $N = 50, 100, 200, 400$ respectively, increasing by close to 1.0 per doubling of N (consistent with $d_s \sim \log_2 N + \text{const}$). The previously reported value $d_s \approx 5.25$ is simply the $N = 100$ instance of this size-dependent quantity, not a universal constant, and we correct that claim here. Establishing whether d_s has a well-defined continuum limit (e.g. via finite-size extrapolation) is left for future work.

For the fluid sector, we perform an explicit, reproducible power-law fit of the near-boundary flow, pooling all three topologies and all four values of N :

$$1 + w_{\text{eff}} = C \left(1 - \frac{\lambda_1}{N}\right)^p, \quad p = 1.11 \pm 0.02, \quad C = 0.575, \quad (8)$$

($R^2 = 0.94$ pooled; per- N fits give $p = 1.12, 1.11, 1.11, 1.10$ for $N = 50, 100, 200, 400$, i.e. stable to within its fit uncertainty). This replaces a previously quoted pair of exponents ($A = 0.0166$, $B = 2.5310$) for which no derivation or code existed in the accompanying repository; we withdraw those numbers and report only the reproducible exponent p above, which does appear stable against changes in N and topology in this diagnostic.

VI. CONCLUSION

We have presented a candidate cosmological toy model in which an effective energy density and equation of state are derived from the spectral partition function of an MI-Laplacian on a relational substrate. The early-time de Sitter fixed point (Eq. (3) at $\alpha \rightarrow 0^+$) is the strongest result in this work: it follows from an exact scaling identity (Sec. II) and is stable against substrate size ($N = 50\text{--}400$) and topology (Sec. V) in the $(\lambda_1/N, w_{\text{eff}})$ projection, though not in the spectral dimension d_s , which we find to scale with N rather than converge to a universal value.

The mechanisms connecting this fixed point to a graceful exit (Eq. (4)) and to a late-time dark-energy-like feature (Eq. (5)) are explicit phenomenological ansätze with stated free parameters, not independently derived substrate dynamics; in this respect the framework relocates, rather than eliminates, the fine-tuning questions it set out to address. A direct fit of the late-time ansatz to the real Pantheon+SH0ES compilation (Sec. IV A) shows it is statistically indistinguishable from flat Λ CDM, with a fractional luminosity-distance deviation between the two best-fit curves of order 1–3% rather than the sub-percent bound previously and incorrectly claimed.

We regard these results as motivation for further work – in particular, deriving $\alpha(t)$ and the late-time re-entanglement feature from an independent equation of motion for the substrate, extending the Pantheon+ fit to use the full systematic covariance matrix, and establishing whether the spectral dimension admits a genuine continuum limit – rather than as a demonstrated replacement for Λ CDM. All code, data-fetching scripts, and figures underlying these results are released alongside the paper for independent verification.

Appendix A: Numerical β -invariance of the equation-of-state identity

Table I reports the output of `code/beta_invariance_break_test.py`, which sweeps β over four orders of magnitude and checks two quantities on the $L = 8$ torus substrate ($N = 64$): the $\alpha = 0$ energy ceiling (Sec. III) and the deviation of the $\alpha = 1$ equation of state from the exact $\beta \rightarrow 0^+$ identity $w_{\text{eff}} = -2/3$ (Eq. (3)).

β	$\rho_{\text{eff}}(\alpha = 0)$	$w_{\text{eff}}(\alpha = 1)$	$ \Delta w_{\text{eff}} $
0.001	64.0000	-0.6667	3.6×10^{-5}
0.01	64.0000	-0.6670	3.7×10^{-4}
0.1	64.0000	-0.6718	5.2×10^{-3}
1.0	64.0000	-0.6947	2.8×10^{-2}
10.0	64.0000	-0.6667	7.0×10^{-3}

TABLE I. β -invariance break-test. The $\alpha = 0$ ceiling is exactly $\rho_{\text{eff}} = N = 64$ for every β tested, as expected since the spectrum is fully degenerate there. The $\alpha = 1$ equation of state deviates from the exact identity by less than 1% (relative) for $\beta \lesssim 0.1$, growing to $\sim 4\%$ (relative) at $\beta = 1.0$ before returning to a very small value at $\beta = 10$ (a coincidence of this particular test point rather than evidence of monotonic behavior). This motivates, but does not by itself fully justify, the use of $\beta = 0.1$ throughout the figures in this paper; a systematic study of the β -dependence of all reported quantities is left for future work.

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